# Prottoy Modak

**CSE Undergraduate**

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## CAREER OBJECTIVE

Seeking a Software Engineer or AI/ML Engineer position where I can apply my skills in deep learning (computer vision, self-supervised learning, NLP) and full-stack web development (Next.js, React, Python, C++) to build reliable, scalable, and practical software solutions and intelligent systems.

## EDUCATION

**BSc in Computer Science & Engineering** — Expected 2026
East West University, Dhaka

**Higher Secondary Certificate (HSC)** — 2019 – 2021
Dhaka Imperial College

**Secondary School Certificate (SSC)** — 2017 – 2019
Rajdhani Ideal School & College

## TECHNICAL SKILLS

**Programming:** C, C++, Python, Java, JavaScript, TypeScript, PHP, SQL

**Web:** HTML, CSS, React, Next.js, Vite, Tailwind CSS, MongoDB

**AI/ML:** Machine Learning, Deep Learning, Computer Vision, NLP, Self-Supervised Learning (PyTorch, Hugging Face Transformers)

**CS Fundamentals:** Data Structures & Algorithms

**Tools:** Git, GitHub, VS Code, PyCharm, Visual Studio, Eclipse

## PROJECTS

### AI / MACHINE LEARNING

**FoodBD Semantic Segmentation** — *PyTorch, DeepLabV3, SegFormer, DINOv2, BYOL, MAE, SimCLR*
*github.com/prottoym/FoodBD-Semantic-Segmentation*

- Compared supervised models on 3,523 Bangladeshi food images (67 classes); DeepLabV3-ResNet50 achieved the best result at 0.5208 mIoU.
- Evaluated four self-supervised methods (SimCLR, BYOL, MAE, DINOv2); DINOv2 reached 0.4359 mIoU using only 697 labels, recovering 90.8% of SegFormer-B0's supervised mIoU with ~3.55× fewer labels.

**Warli Art Classification** — *PyTorch, Swin-Tiny, BYOL, DINO*
*github.com/prottoym/Warli-Art-Classification*

- Trained Swin-Tiny on 7,535 images across 14 Warli art classes, reaching 99.60% supervised accuracy.
- Applied self-supervised BYOL and DINO with linear-probe and k-NN evaluation; retained ~88% accuracy using only 10% of labels (~90% less labeling effort).

**A Benchmark for Bangla Task-Oriented Dialogue Understanding (Ongoing)** — *NLP, BERT, mBERT, BanglaBERT*

- Developing a task-oriented dialogue system for Bangla, English, and Banglish covering 9 domains and 64 intents.
- Fine-tuning and comparing English BERT, Multilingual BERT, and BanglaBERT for intent classification and slot filling, evaluated with accuracy, Macro F1, and Weighted F1.

### SOFTWARE DEVELOPMENT

**BazarDor — Bangla Market Price Tracker** — *Next.js, TypeScript, Tailwind CSS, MongoDB, Better Auth*
*github.com/prottoym/Bazar-Dor* | *Live Demo*

- Built and deployed a responsive Bangla web app showing daily prices of essential goods, with market-wise comparison and daily price risers/fallers.
- Implemented secure authentication (email/password, Google, GitHub) with protected routes, category sorting, and a live price ticker.

**FitLog — Workout Planner** — *Next.js, Tailwind CSS, REST API*
*github.com/prottoym/Fit-Log*

- Developed a workout library and daily planner that fetches data from a REST API, with sorting, detail pages, and live plan statistics (duration, calories).
- Added save-for-later and mark-as-done features with data persisted via localStorage and toast notifications.

**DevStack — Tech Stack Builder** — *React, TypeScript, Vite, Tailwind CSS*
*github.com/prottoym/b-14-assignment-05*

- Built a responsive React app where users browse 15 web technologies loaded dynamically from JSON and add favorites to a personal "Your Stack" panel.
- Implemented duplicate-add prevention, toast notifications (React-Toastify), a mobile hamburger menu, and an adaptive 1/2/3-column grid.

**Algorithm Visualizer** — *React, JavaScript, Vite, Tailwind CSS, Framer Motion*
*github.com/prottoym/Algorithm-Visualizer*

- Built an interactive platform visualizing 8 sorting, searching, and pattern-matching algorithms with step-by-step animations, execution logs, and complexity breakdowns.
- Designed reusable React components and state-driven controls for custom input, animation speed, random test cases, and execution interruption.

## CERTIFICATIONS

Kick Start to Robotics 5.0 — Robotics Training Program
JavaScript Basics — University of California, Davis | Coursera (Certificate)
Web Development Course — Programming Hero *(In Progress)*

## EXTRACURRICULAR ACTIVITIES

Volunteer, East West University Robotics Club — Assisted in organizing robotics events and club activities.